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## PART 1 — THE 30 QUESTIONS YOU MUST KNOW

If you have very little time, **memorize these first.**

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### Q1. Give us a brief introduction to your project.

#### Answer

Our project is an AI-assisted multimodal logistics optimization platform designed primarily for MSME and agricultural logistics.

It solves two different types of logistics risks. For perishable goods such as vaccines and organic produce, it estimates spoilage risk based on temperature and transit/dwell time using Q10-based decay modeling. For non-perishable goods, it estimates economic risk due to delivery delays and SLA penalties.

We then combine freight cost and expected loss into a multi-objective optimization function and use Google OR-Tools CP-SAT to generate different consolidation and routing plans.

The system considers a 20-city network containing multimodal hubs and satellite towns and provides three plans: cheapest, fastest/lowest-risk, and balanced.

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### Q2. What problem are you actually solving?

#### Answer

The problem is that logistics decisions are usually not simply about finding the cheapest route.

A cheap route can increase transit time, dwell time, temperature exposure and consequently spoilage or delay-related financial losses.

Our system tries to find a better trade-off between **freight cost and expected loss**, while also considering multimodal transportation and consolidation.

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### Q3. Why is this problem important?

#### Answer

MSMEs and agricultural shippers often have limited logistics budgets, while perishable goods can lose significant value if transportation takes too long or temperature conditions are violated.

Therefore, minimizing only transportation cost isn't sufficient. We need to consider the **total economic consequence of the logistics decision**, including potential cargo loss and penalties.

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### Q4. What are the two types of cargo in your system?

#### Answer

There are:

### **Class A — Perishable cargo**

Examples:

- vaccines
- organic produce

Risk comes mainly from:

**temperature × time**

### **Class B — Non-perishable cargo**

Risk comes primarily from:

**delivery delay × contractual/economic penalty**

This separation is explicitly part of your model.

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## **Q5. Why separate Class A and Class B?**

### **Answer**

Because their failure mechanisms are fundamentally different.

For a vaccine, temperature exposure can directly compromise the cargo even if the shipment isn't technically late.

For a non-perishable shipment, temperature isn't necessarily the dominant concern; instead, missing the SLA can cause financial penalties.

Therefore, using one generic risk formula for both would be inappropriate.

This is a **very good answer** because it shows that your classification isn't arbitrary.

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## **Q6. What is multimodal transportation?**

### **Answer**

Multimodal transportation means using more than one mode of transport in the same logistics journey.

For example:

**Local road → multimodal hub → trunk rail → multimodal hub → road → destination.**

In our network, satellite towns are road-connected while major hubs provide road and trunk-rail connectivity.

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## **Q7. Why use both road and rail?**

### Answer

Road provides flexibility and direct access to smaller destinations, whereas rail can provide economical bulk transportation over longer trunk corridors.

Therefore, the optimizer can consolidate shipments onto rail where that makes economic sense and use road for first-mile and last-mile connectivity.

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### Q8. What is consolidation?

#### Answer

Consolidation means combining compatible shipments traveling along the same corridor or route into a common transportation movement.

Instead of sending multiple partially filled vehicles independently, compatible shipments can be grouped, improving vehicle utilization and potentially reducing freight cost.

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### Q9. What is the objective function?

Your README gives:

$$\min(\alpha \times \text{freight\_cost} + \beta \times \text{expected\_loss})$$

where:

$$\alpha + \beta = 1$$

#### Answer

The optimizer minimizes a weighted combination of freight cost and expected loss.

Alpha represents how much we care about transportation cost, while beta represents how much we care about risk-related expected loss.

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### Q10. Why must $\alpha + \beta = 1$ ?

#### Answer

It normalizes the two weights into a relative trade-off.

For example, if  $\alpha = 0.9$  and  $\beta = 0.1$ , the optimization is strongly cost-oriented.

If  $\alpha = 0.2$  and  $\beta = 0.8$ , the optimizer prioritizes risk reduction.

The actual importance comes from the relative values of the weights.

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### Q11. Explain your three optimization scenarios.

#### Answer

Your README defines:

| Scenario            | Cost | Risk |
|---------------------|------|------|
| Cheapest            | 90%  | 10%  |
| Fastest/Lowest-Risk | 20%  | 80%  |
| Balanced            | 55%  | 45%  |

**Say:**

We generate three different decision preferences.

**Cheapest:** prioritizes freight cost and therefore favors economical consolidation such as rail.

**Fastest/lowest-risk:** prioritizes reducing transit and dwell time, so it tends to favor more direct road dispatch.

**Balanced:** provides a compromise between the two.

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## Q12. What is Pareto optimality?

**Answer**

A solution is Pareto-optimal if we cannot improve one objective without worsening another.

For example, suppose one plan costs less but has higher expected loss, while another costs more but has significantly lower expected loss.

Neither necessarily dominates the other.

These trade-off solutions form the Pareto frontier.

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## Q13. Why not simply choose the cheapest route?

**Answer**

This is a **very important question**.

Because the cheapest route can actually have the highest total economic cost.

Suppose a ₹1 lakh saving in freight causes ₹5 lakh worth of expected cargo loss. Then the apparently cheaper route is economically worse.

That's why we optimize freight cost together with expected loss.

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## Q14. What is expected loss?

**Answer**

For Class A:

$$\text{Expected Loss} = \text{Risk Score} \times \text{Cargo Value}$$

For Class B:

$$\text{Expected Loss} = \text{Risk Score} \times \text{Cargo Value} \times \text{Delay Penalty Rate}$$

**Say:**

Expected loss converts a risk probability or risk score into a monetary quantity.

This allows us to compare logistics cost and potential cargo loss within the same optimization objective.

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#### **Q15. What is Q10?**

**Answer**

Q10 represents the temperature sensitivity of a biological or chemical process.

It describes approximately how much the reaction or degradation rate changes when temperature increases by 10°C.

In our implementation, different cargo types have different Q10 factors.

Your README uses:

- Vaccines: **2.5**
  - Grapes: **2.2**
- 

#### **Q16. Explain this formula.**

$$\text{fraction\_consumed} = \frac{\text{transit\_time} + \text{dwell\_time}}{\text{base\_shelf\_life}} \times Q_{10}^{\frac{T_{\text{actual}} - T_{\text{optimal}}}{10}}$$

**Answer**

Break it into two components.

**First:**

$$\frac{\text{transit} + \text{dwell}}{\text{shelf life}}$$

represents how much of the nominal shelf life has been consumed by time.

**Second:**

$$Q_{10}^{(T_{\text{actual}} - T_{\text{optimal}})/10}$$

represents the temperature acceleration factor.

Then:

We multiply these two effects to estimate the fraction of usable shelf life consumed.

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**Q17. What happens if actual temperature equals optimal temperature?**

Then:

$$T_{actual} - T_{optimal} = 0$$

Therefore:

$$Q_{10}^0 = 1$$

So:

$$fraction\ consumed = \frac{transit + dwell}{shelf\ life}$$

**Excellent interview answer:**

At the optimal temperature, there is no additional temperature acceleration according to the model.

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**Q18. What happens if temperature increases by 10°C?**

If:

$$T_{actual} = T_{optimal} + 10$$

then:

$$Q_{10}^1 = Q_{10}$$

So for vaccines:

$$2.5 \times$$

the degradation factor becomes **2.5 times** the baseline.

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**Q19. Why is Q10 different for vaccines and grapes?**

**Answer**

Different biological and pharmaceutical products have different temperature sensitivities.

Therefore, we don't use one universal Q10 value. The README defines separate factors for vaccines and grapes.

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**Q20. What is the risk score?**

Your formula is:

$$risk\_score = \min(1, \max(0, fraction\_consumed))$$

**Answer**

It represents normalized risk between 0 and 1.

Zero means essentially no modeled loss, while 1 represents the maximum modeled loss.

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**Q21. Why clamp the risk score between 0 and 1?****Answer**

Because a risk score is normalized to represent 0–100%.

Mathematical calculations could potentially produce a value greater than 1, so we cap it at 1. Similarly, negative values are prevented by the lower bound of 0.

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**Q22. What is a hard breach?****Answer**

Your condition is:

$$breached = \neg(T_{min} \leq T_{actual} \leq T_{max})$$

**Answer**

A hard temperature breach occurs when the actual temperature goes outside the permitted temperature band.

For vaccines:

**2°C–8°C**

For grapes:

**4°C–12°C**

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**Q23. Why does a vaccine get 100% risk after a hard breach?****Answer**

In our current model, medical vaccines use a hard-breach override because temperature excursions can make the shipment unusable from a cold-chain compliance perspective.

Therefore, if the hard-breach condition is triggered, the risk score becomes 1.0.

Your README explicitly specifies this behavior.

 **If challenged:**

Judge:

“But is every vaccine necessarily destroyed immediately after going outside 2–8°C?”

Don't say yes.

Say:

“Not necessarily. That depends on the specific vaccine, excursion duration and validated stability data. Our current implementation uses a conservative hard-breach rule for the modeled medical-

cargo scenario. A production version would ideally incorporate product-specific excursion stability curves.”

**This is a much stronger answer.**

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#### **Q24. What is dwell time?**

##### **Answer**

Dwell time is the time a shipment spends waiting or being handled at an intermediate location rather than actively moving.

For perishables, dwell time matters because the cargo continues aging even when it isn't traveling.

This is an important insight.

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#### **Q25. Why include dwell time in the formula?**

##### **Answer**

Because:

$$Total\ exposure = Transit\ time + Dwell\ time$$

A truck traveling for 20 hours and waiting for 10 hours effectively exposes the cargo for **30 hours**.

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#### **Q26. What is the Class B risk formula?**

$$risk\_score = delay\_probability$$

and

$$expected\ loss = delay\ probability \times cargo\ value \times penalty\ rate$$

##### **Say:**

For non-perishable cargo, the important event is an SLA delay. We estimate the probability of delay and multiply it by the shipment value and the contractual penalty fraction.

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#### **Q27. Why is delay penalty rate dimensionless?**

##### **Answer**

Suppose:

- Cargo value = ₹10 lakh
- Delay probability = 0.2
- Penalty rate = 0.1

Then:



$$10,00,000 \times 0.2 \times 0.1 = ₹20,000$$

The penalty rate is a fraction rather than a monetary amount.

Your README specifically constrains it to:

$$0 \leq \text{rate} \leq 1$$

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#### **Q28. What is CP-SAT?**

##### **Answer**

CP-SAT is Google's constraint programming and satisfiability-based optimization solver available through OR-Tools.

We use it to solve the combinatorial optimization problem of assigning shipments to different transportation and consolidation choices while respecting constraints.

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#### **Q29. Why CP-SAT instead of a normal shortest-path algorithm?**

##### **Answer**

This is **VERY likely**.

A shortest-path algorithm such as Dijkstra primarily solves a graph path problem.

Our problem is more complex because we have multiple shipments, consolidation decisions, transportation modes, cost, risk, capacity and potentially scheduling constraints.

Therefore, we're solving an optimization problem rather than simply finding one shortest path.

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#### **Q30. Where is the AI?**

##### **Best answer**

The intelligence in the current system is primarily optimization intelligence rather than a trained machine-learning model.

We use deterministic physics-based risk modeling and constraint optimization through OR-Tools to make decisions.

We intentionally chose this approach because logistics decisions require explainability and hard constraints. A future version can incorporate ML models for components such as delay probability, demand forecasting and temperature prediction.

##### **DO NOT SAY:**

“CP-SAT is machine learning.”

It isn't.

If a judge knows ML, that answer will hurt you.

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## **PART 2 — BASIC QUESTIONS**

### **Q31. What technologies did you use?**

According to the README:

#### **Backend**

- Python
- FastAPI
- Pydantic
- OR-Tools
- pytest

#### **Frontend**

- React
- TypeScript
- Axios
- Tailwind CSS
- Vite

#### **Infrastructure**

- Docker
- Docker Compose
- GitHub Actions / CI

#### **External routing**

- OSRM

The backend and frontend structure is explicitly documented in the README.

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### **Q32. Why FastAPI?**

#### **Answer**

FastAPI provides a lightweight Python framework for building REST APIs, has automatic API documentation through OpenAPI, supports Pydantic validation and works well for computational backend services.

Your API documentation is exposed through /docs.

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### **Q33. What is REST API?**

**Answer**

REST API is an interface through which frontend or external clients communicate with backend services using HTTP methods such as GET and POST.

Examples from our project:

GET /health

GET /api/v1/checkpoints

GET /api/v1/seed-demo

POST /api/v1/plan

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**Q34. Why POST for /plan?****Answer**

Because we're sending shipment information and planning parameters to the server in the request body and asking the backend to compute a new plan.

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**Q35. Why GET for checkpoints?****Answer**

Because we're simply retrieving existing network information rather than creating or modifying a resource.

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**Q36. What is Pydantic?****Answer**

Pydantic is used for data validation and structured data models in Python.

In our system it helps validate API request and response schemas and shipment-related data.

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**Q37. Why TypeScript instead of plain JavaScript?****Answer**

TypeScript provides static typing, which reduces frontend errors and makes it easier to maintain interfaces that mirror the backend data models.

Your README specifically mentions TypeScript interfaces mirroring backend models.

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**Q38. Why React?****Answer**

React allows us to build the dashboard as reusable components and manage different views such as maps, cards, tabs and shipment information.

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### **Q39. Why Docker?**

#### **Answer**

Docker makes the application environment reproducible.

Instead of requiring someone to manually configure Python, Node, dependencies and services, we can package the application and run the complete system using Docker Compose.

The README provides:

```
docker compose up --build
```

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### **Q40. What is OSRM?**

#### **Answer**

OSRM stands for Open Source Routing Machine.

It is used for road-route calculations. Our backend contains an OSRM wrapper with timeout, backoff and fallback handling.

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### **Q41. Why do you need OSRM if you already have OR-Tools?**

#### **Answer**

Excellent question.

They solve different problems.

OSRM provides road-network routing information such as road distance and route information.

OR-Tools performs the higher-level optimization and decides among different transportation and consolidation choices.

So OSRM can be considered a routing-data provider, while CP-SAT is the decision optimizer.

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### **Q42. What is the backend pipeline?**

Your architecture suggests:

**Input → ingestion/validation → classification → route generation → risk calculation → consolidation → optimization → output**

The relevant modules are explicitly separated in the README.

**Say:**

The shipment enters through the API, gets validated, classified as Class A or B, candidate modal routes are generated, risk is calculated, shipments can be grouped for consolidation, and then CP-SAT selects the optimized plans.

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### **PART 3 — QUESTIONS ABOUT YOUR ARCHITECTURE**

#### **Q43. Explain your architecture.**

##### **Answer**

Think:

Frontend



FastAPI



Input Validation



Classification



Route Generation



Risk Engine



Consolidation



CP-SAT Optimizer



3 Plans



Frontend Dashboard

You should be able to draw this on a whiteboard.

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#### **Q44. Why separate the risk engine from the optimizer?**

##### **Answer**

Separation of concerns.

The risk engine is responsible for calculating risk consistently according to the mathematical model.

The optimizer shouldn't independently implement risk calculations because that could lead to inconsistent results.

Having a single risk module makes the model easier to test and maintain.

Your README actually describes risk.py as containing the deterministic physics and **single-source formulas**.

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#### **Q45. What is separation of concerns?**

##### **Answer**

It means each software component has a well-defined responsibility.

For example:

risk.py → risk calculation

road\_route.py → road routing

optimization\_engine.py → optimization

schemas.py → API data structures.

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#### **Q46. Why have an orchestrator?**

##### **Answer**

The orchestrator coordinates the different components.

Instead of the API directly controlling classification, routing, risk calculation and optimization individually, the orchestrator coordinates the entire pipeline.

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#### **Q47. What is repository abstraction?**

##### **Answer**

Your README includes:

shipment\_repository.py — Data-access abstraction layer

##### **Answer**

It separates data access from business logic.

This means the optimization engine doesn't need to know whether shipment data comes from a database, file, API or another source.

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#### **Q48. Why have separate models for RoadLeg and RailLeg?**

##### **Answer**

Road and rail have different properties and constraints.

Separating them makes the system more explicit and easier to extend with mode-specific attributes.

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## **PART 4 — NETWORK QUESTIONS**

**Q49. How many cities are in your network?**

**20 cities.**

- 5 multimodal hubs
  - 15 satellite towns
- 

**Q50. How many hubs?**

**5.**

They are:

1. Indranagar Junction
  2. Himkot
  3. Suryapatan
  4. Chandanpalli
  5. Meghdoot
- 

**Q51. How many satellite towns?**

**15.**

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**Q52. Why hubs?**

**Answer**

Hubs allow shipments from multiple smaller locations to converge and be consolidated before being transported through high-capacity trunk corridors.

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**Q53. Why are satellite towns road-only?**

**Answer**

The modeled network assumes smaller towns have road connectivity to their nearest logistics hubs, while trunk rail infrastructure connects the major hubs.

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**Q54. What are the five trunk rail corridors?**

Know these.

1. Indranagar ↔ Himkot
  2. Indranagar ↔ Suryapatan
  3. Indranagar ↔ Chandanpalli
  4. Indranagar ↔ Meghdoot
  5. Suryapatan ↔ Chandanpalli
- 

**Q55. Why is there a Suryapatan–Chandanpalli bypass?**

**Answer**

It provides a direct west-south connection and potentially avoids routing through the central hub.

This can reduce distance, travel time or congestion depending on the shipment.

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## **PART 5 — TOUGH MATHEMATICAL QUESTIONS**

These are the questions that can separate teams.

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**Q56. If  $Q_{10} = 2.5$ , what happens when temperature rises 20°C above optimal?**

Formula:

$$Q_{10}^{20/10}$$

Therefore:

$$2.5^2 = 6.25$$

So the modeled degradation acceleration factor is:

**6.25×**

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**Q57. If temperature is 10°C below optimal?**

Mathematically:

$$Q_{10}^{-1} = \frac{1}{Q_{10}}$$

For vaccine:

$$\frac{1}{2.5} = 0.4$$

So the modeled rate factor becomes **0.4×** baseline.

**But careful:**



The safe temperature-band logic still applies separately.

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**Q58. Suppose vaccine:**

- transit = 12 h
- dwell = 6 h
- shelf life = 48 h
- optimal temperature = 5°C
- actual temperature = 5°C

Calculate risk.

Time:

$$12 + 6 = 18$$

Fraction:

$$18/48 = 0.375$$

Temperature factor:

$$2.5^0 = 1$$

Therefore:

$$risk = 0.375$$

So:

**37.5% modeled risk.**

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**Q59. Same shipment at 15°C?**

Difference:

$$15 - 5 = 10$$

Temperature factor:

$$2.5^1 = 2.5$$

Fraction:

$$0.375 \times 2.5 = 0.9375$$

Risk:

**93.75%**

And because 15°C is outside the vaccine's safe band of 2–8°C, the hard-breach rule would make the medical-cargo risk **1.0** in the current implementation.

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**Q60. What if risk becomes 1.5?**

It gets clamped:

$$\min(1, \max(0, 1.5)) = 1$$

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**Q61. What if risk is -0.2?**

$$\min(1, \max(0, -0.2)) = 0$$

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**Q62. Why use min and max instead of if-else?**

**Answer**

Both approaches could enforce the bounds. The min-max formulation is a concise mathematical representation of clipping the risk score into the valid  $[0,1]$  interval.

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**Q63. What does  $\alpha = 0.9$ ,  $\beta = 0.1$  mean?**

**Answer**

The optimizer is heavily cost-oriented.

A change in freight cost has nine times the objective weight of an equivalent normalized change in expected loss.

**Important caveat:**

Only say “**equivalent normalized change**” because cost and loss may have different scales.

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**Q64. What is the problem with directly adding freight cost and expected loss?**

This is a **very good tough question**.

Suppose:

- freight cost = ₹10,000
- expected loss = ₹10,00,000

Then adding them directly means expected loss dominates numerically.

**Answer:**

Before combining objectives, we need to ensure their scales are compatible. The README defines the weighted objective, but in a production implementation I'd explicitly normalize or scale the objective terms so that the weights  $\alpha$  and  $\beta$  have meaningful comparative influence.

**This is a sophisticated answer.**

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**Q65. Is your objective truly multi-objective optimization?****Answer**

Conceptually yes, because we're considering two competing objectives: freight cost and expected loss.

The current implementation scalarizes those objectives into a weighted single objective:

$$\alpha C + \beta L$$

and runs different weight combinations to expose different trade-offs.

That is more technically accurate than simply saying “CP-SAT does Pareto optimization.”

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**Q66. Does CP-SAT guarantee the globally optimal solution?****Answer****Careful.**

CP-SAT is an exact optimization solver for the modeled integer/constraint problem, but practical performance depends on the model size, time limit and solver configuration.

For our current 20-city benchmark, we obtain solutions within the required latency budget.

Don't casually say:

“It always gives the globally optimal solution instantly.”

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**PART 6 — PERFORMANCE QUESTIONS****Q67. What is your latency?**

README:

**p50 = 46.58 ms**

**p95 = 55.16 ms**

Target:

**p95 < 2000 ms**

**Answer**

In our current benchmark, p50 is approximately 46.58 ms and p95 is approximately 55.16 ms, against a target of below 2000 ms.

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**Q68. What is p50?****Answer**

p50 is the median latency.

Half of the requests finish faster than that value and half take longer.

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#### **Q69. What is p95?**

##### **Answer**

p95 means 95% of measured requests complete within that latency, while approximately 5% take longer.

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#### **Q70. Why is p95 more useful than average?**

##### **Answer**

Average can hide occasional very slow requests.

p95 gives a better idea of the user experience for high-percentile latency and system reliability.

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#### **Q71. How did you test latency?**

README says:

CP-SAT Optimization Latency Stress Test

and mentions a 20-solve benchmark.

##### **Answer:**

We use a dedicated stress-test script that runs the CP-SAT optimization repeatedly and measures latency percentiles.

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#### **Q72. Is 55 ms really production latency?**

##### **Correct answer:**

It's the measured benchmark for our current modeled workload and environment.

It shouldn't automatically be interpreted as guaranteed production latency at national scale.

Real production performance would depend on shipment volume, network size, concurrent requests, external routing calls, infrastructure and solver configuration.

**This answer will impress technically competent judges.**

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## **PART 7 — TESTING QUESTIONS**

#### **Q73. How many tests?**

**31 tests**

with:

**92% coverage**

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**Q74. What is unit testing?**

**Answer**

Testing an individual component or function independently to verify that it behaves correctly for expected and edge-case inputs.

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**Q75. What is end-to-end testing?**

**Answer**

Testing the complete workflow from input through processing to final output.

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**Q76. Why 92% coverage instead of 100%?**

**Answer**

Coverage is a useful indicator, but 100% coverage doesn't automatically mean a system is bug-free. We prioritize meaningful tests for critical business logic, edge cases and end-to-end workflows.

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**Q77. What should you test in the risk engine?**

Say:

- normal temperature
  - temperature below safe range
  - temperature above safe range
  - zero transit
  - high dwell time
  - risk > 1
  - negative/invalid inputs
  - different cargo types
  - different Q10 values
  - hard-breach behavior
- 

**Q78. What should happen if invalid temperature is provided?**

## Answer

The API/input validation layer should reject invalid data before it reaches the optimization engine, rather than allowing invalid values to silently affect the calculation.

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## PART 8 — API QUESTIONS

### Q79. What does /health do?

It provides service health and version status.

README:

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### Q80. What does /seed-demo do?

It returns the six seed demonstration shipments.

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### Q81. What does /plan do?

It accepts shipment IDs and planning parameters and generates the three optimization plans.

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### Q82. What does this input mean?

```
{  
  "shipment_ids": [  
    "SHP-001",  
    "SHP-002",  
    "SHP-003",  
    "SHP-004",  
    "SHP-005",  
    "SHP-006"  
  ],  
  "simulated_temp_c": 5.0  
}
```

## Answer

It tells the planning API which shipments should be considered and supplies the simulated temperature used for the risk calculation.

The exact sample is in your README.

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## PART 9 — “WHY DID YOU USE X?” QUESTIONS

These are **extremely common in SIH**.

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### Q83. Why Python?

Strong ecosystem for optimization, scientific computation, APIs and rapid development.

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### Q84. Why React?

Component-based architecture makes it suitable for building an interactive logistics dashboard.

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### Q85. Why FastAPI?

Fast development, validation through Pydantic and automatic API documentation.

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### Q86. Why OR-Tools?

It provides mature optimization algorithms and CP-SAT is well suited to constraint-heavy combinatorial problems.

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### Q87. Why CP-SAT instead of genetic algorithm?

#### Strong answer:

Genetic algorithms are useful for large heuristic search spaces, but they are stochastic and don't naturally provide the same constraint-programming formulation.

Our problem contains explicit discrete decisions and hard constraints, so CP-SAT provides a more interpretable and deterministic optimization framework.

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### Q88. Why not reinforcement learning?

#### Answer

Reinforcement learning would require defining states, actions, rewards and training through sufficient experience.

Our current problem already has explicit business constraints and mathematical cost/risk functions, so constraint optimization is more directly appropriate.

RL could become useful later for dynamic dispatch or adaptive decisions based on historical operational data.

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### Q89. Why not a neural network?

## **Answer**

A neural network is not inherently necessary for every optimization problem.

If the relationship can be expressed using known physics and explicit constraints, a deterministic model is more interpretable and easier to validate.

Neural networks could be useful for components where we have historical data, such as predicting delay probability or temperature excursions.

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## **Q90. Why call it AI then?**

### **Best answer:**

We use “AI” in the broader decision-intelligence sense because the system automatically evaluates alternatives and performs constrained optimization.

However, the current implementation is not primarily a machine-learning model. Its core consists of deterministic risk models and mathematical optimization.

We intentionally prioritized explainability and constraint satisfaction, and ML can be incorporated where historical data provides predictive value.

**This is probably one of the most important answers for your team.**

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## **PART 10 — REAL-WORLD CHALLENGE QUESTIONS**

These are the questions judges ask when they want to know whether this is merely a college project.

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## **Q91. What if the road is blocked?**

### **Answer**

The route layer should provide alternative routes or fallbacks, and the optimizer can evaluate alternative candidate routes.

Our road routing wrapper already includes timeout/backoff and fallback handling.

README supports the timeout/backoff/fallback part.

### **If asked whether you currently model real-time road closures:**

Not explicitly in the current README-described model. That would be a valuable extension using live traffic/incidents data.

**Don't pretend you implemented something you didn't.**

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## **Q92. What if OSRM is unavailable?**

### **Answer**



Our OSRM wrapper has timeout, backoff and fallback handling.

The important architectural principle is that failure of an external routing dependency shouldn't crash the entire planning service.

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### **Q93. What if rail is delayed?**

#### **Answer**

Rail delay should contribute to the shipment's transit/dwell exposure and therefore increase expected loss for perishable cargo or delay probability for Class B.

A production system would ideally use historical and real-time rail reliability data to update those probabilities.

---

### **Q94. What if temperature changes during transportation?**

This is a **very important weakness to understand**.

Your sample API currently provides:

"simulated\_temp\_c": 5.0

#### **Judge:**

"But real temperature isn't constant, right?"

#### **Answer:**

Correct. The current model uses a temperature value as an input to demonstrate the risk calculation.

A production cold-chain implementation should use time-series IoT sensor data and calculate cumulative thermal exposure over time.

That would make the model more realistic than assuming one constant temperature.

This is exactly the kind of answer that shows maturity.

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### **Q95. Can your system use IoT sensors?**

#### **Answer**

Yes, architecturally the risk engine can consume temperature measurements from IoT devices.

Instead of manually supplying a simulated temperature, an IoT gateway could send timestamped temperature readings, which could then feed the cold-chain risk engine.

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### **Q96. What sensors would you use?**

You can say:

- temperature sensor

- humidity sensor where relevant
- GPS
- accelerometer/shock sensor where relevant
- door/open sensor
- battery monitoring for the device

But **don't claim these are already implemented unless they are in the actual code.**

---

#### **Q97. What happens if the IoT sensor fails?**

##### **Answer**

We should detect missing/stale sensor data and fall back to a conservative risk assumption or historical estimate, while flagging the shipment for attention.

The exact fallback policy would depend on the cargo's criticality.

---

#### **Q98. What if the temperature sensor gives an incorrect reading?**

##### **Answer**

Sensor validation is necessary.

We can apply range checks, temporal consistency checks, sensor-health monitoring and anomaly detection before using measurements in the risk model.

---

#### **Q99. What if there is no historical data?**

##### **Answer**

Our current deterministic model doesn't require a large historical dataset to calculate the physics-based risk.

That's actually one advantage of the approach.

Historical data becomes particularly important when we want to learn empirical delay probabilities, demand patterns or sensor anomalies.

---

### **PART 11 — MACHINE LEARNING QUESTIONS**

If SIH judges see “AI” anywhere, **expect these.**

---

#### **Q100. What is supervised learning?**

Learning a mapping from input features to known target labels/values using labeled training data.

---

**Q101. What ML model would you add to this project?**

A very strong answer:

I would add an ML model for **delay probability prediction**.

Features could include:

- route
- transportation mode
- historical travel time
- day of week
- season
- weather
- congestion
- hub
- historical carrier performance

The model could output:

$$P(\text{delay})$$

which can directly feed the Class B expected-loss calculation.

---

**Q102. Would you replace CP-SAT with ML?**

**Answer**

No.

ML and optimization solve different parts of the problem.

ML can predict uncertain quantities such as delay probability, demand or ETA.

CP-SAT can then make the final constrained decision using those predictions.

This is **excellent architecture**:

Historical Data



ML



Delay Probability



Risk Model



CP-SAT



Optimal Logistics Plan

---

### Q103. What ML model could predict delay?

Possible answers:

- XGBoost
- LightGBM
- Random Forest
- Logistic Regression

For probability prediction, you could say:

Initially I'd benchmark logistic regression against tree-based models such as XGBoost, because the dataset may contain nonlinear interactions between route, weather, congestion and logistics conditions.

---

### Q104. What would be your target variable?

For delay prediction:

$$y = \begin{cases} 1 & \text{if shipment misses SLA} \\ 0 & \text{otherwise} \end{cases}$$

---

### Q105. What evaluation metrics?

For delay classification:

- Precision
- Recall
- F1-score
- ROC-AUC
- PR-AUC
- calibration

#### Particularly good:

Since the output is a probability used for expected loss, probability calibration is important, not just classification accuracy.

That is a **very strong DS/AI answer**.

---

**Q106. Why not use accuracy?**

If only a small percentage of shipments are delayed, a model predicting “no delay” every time could achieve high accuracy while being useless.

Therefore, precision, recall, F1, PR-AUC and calibration can be more meaningful.

---

**PART 12 — SECURITY QUESTIONS**

**Q107. Is your API secure?**

Don't lie.

If you haven't implemented authentication:

The current prototype focuses primarily on the optimization workflow. Production deployment would require authentication, authorization, rate limiting, encrypted communication and proper secrets management.

---

**Q108. What sensitive information could logistics systems contain?**

Potentially:

- shipment details
- cargo value
- business information
- customer information
- location
- vehicle information

Therefore:

Access control and encryption are important in production.

---

**Q109. How would you protect the API?**

Mention:

- HTTPS
- authentication
- authorization
- API rate limiting

- input validation
  - secret management
  - logging/auditing
  - database encryption where appropriate
- 

## **PART 13 — SCALABILITY QUESTIONS**

### **Q110. Your network has only 20 cities. What about 20,000?**

This is a **killer question**.

**Answer:**

The current benchmark is explicitly for a 20-city network.

Scaling to thousands of locations would require hierarchical decomposition rather than blindly solving the entire problem as one monolithic CP-SAT model.

We could partition the network geographically, precompute candidate routes, use hub-level optimization and solve smaller subproblems.

We could also use caching and asynchronous execution.

---

### **Q111. What is the biggest scalability challenge?**

**Answer:**

Combinatorial explosion.

As shipments, routes, vehicles, time windows and consolidation possibilities increase, the number of possible decisions grows rapidly.

Therefore, candidate generation, pruning, decomposition and solver configuration become important.

---

### **Q112. How would you handle 1 million shipments?**

**Answer**

I wouldn't put one million shipments directly into a single optimization model.

I'd use hierarchical planning:

**Shipment clustering → corridor assignment → hub-level optimization → detailed routing → dispatch scheduling.**

I'd also batch planning and use rolling-horizon optimization.

---

### **Q113. What is rolling-horizon optimization?**

Instead of planning the entire future once, the system repeatedly optimizes over a moving future window.

As new information arrives, the plan can be updated.

This is very useful for real logistics.

---

## **PART 14 — BUSINESS QUESTIONS**

### **Q114. Who is your target user?**

From README:

MSME and agricultural logistics shippers across India.

You can expand:

- agricultural producers
  - MSMEs
  - logistics aggregators
  - cold-chain operators
  - pharmaceutical distributors
  - potentially government logistics operations
- 

### **Q115. How does this save money?**

#### **Answer**

Through consolidation and multimodal route selection, the system can reduce unnecessary freight expenditure.

At the same time, by accounting for expected cargo loss, it avoids choosing routes that look cheap but have excessive risk.

---

### **Q116. What is your value proposition?**

Say:

**Don't optimize transportation cost alone; optimize total expected economic impact.**

That's a great one-liner.

---

### **Q117. Why would a logistics company use your system?**

Because it gives decision-makers multiple trade-off options instead of one black-box route.

They can choose:

- cost-focused
- risk-focused
- balanced

depending on the shipment.

---

#### **Q118. What's your competitive advantage?**

##### **Answer**

The key combination is:

**multimodal consolidation + cold-chain physics + economic delay risk + constraint optimization**

rather than treating routing as only a shortest-path problem.

---

#### **PART 15 — TRICK QUESTIONS**

These are the ones I would REALLY practice.

---

#### **Q119. "Is Q10 machine learning?"**

##### **Answer:**

No. Q10 is a temperature sensitivity parameter used in a mathematical degradation model.

---

#### **Q120. "Is OR-Tools AI?"**

##### **Answer:**

OR-Tools itself is an optimization toolkit, not machine learning.

In our project, it provides the optimization intelligence.

---

#### **Q121. "So your project doesn't actually use AI?"**

##### **Answer:**

The current implementation's intelligence comes primarily from mathematical risk modeling and constraint optimization rather than trained ML.

I would describe it more precisely as an AI-assisted or decision-intelligence system rather than claiming that CP-SAT itself is machine learning.

**DO NOT get defensive.**

---

#### **Q122. "Why are you calling this a prediction engine if there's no ML?"**



**Answer:**

Because the system predicts modeled spoilage/economic risk from the shipment's current parameters.

However, I agree that “prediction” here refers to model-based risk estimation rather than statistical ML prediction.

An ML-based delay prediction module is a natural next step.

---

**Q123. “Your formula assumes constant temperature. Isn't that unrealistic?”**

**Answer:**

Yes, a constant temperature is a simplification.

The current implementation demonstrates the deterministic risk framework. A production cold-chain version should integrate timestamped IoT temperature data and compute cumulative thermal exposure.

---

**Q124. “What if the shipment temperature fluctuates?”**

**Answer:**

Then we should divide the journey into time intervals and calculate temperature-dependent degradation for each interval.

Conceptually:

$$Risk\ Exposure = \sum_i \Delta t_i \times Q_{10}^{(T_i - T_{optimal})/10}$$

followed by appropriate normalization against shelf life.

**This is an excellent answer.**

---

**Q125. “Why use a hard breach for vaccines?”**

**Answer:**

The current model intentionally uses a conservative hard-breach rule for medical cargo.

But actual pharmaceutical stability depends on the product and excursion duration, so a production implementation should use validated product-specific stability information.

---

**Q126. “Why are your city names fictional?”**

Your README uses names such as:

- Indranagar Junction

- Himkot
- Suryapatan
- etc.

**Answer:**

The current network is a controlled demonstration network designed to validate the optimization architecture and benchmark the solver.

The architecture is designed so that the network data can later be replaced with real Indian logistics nodes.

**Don't pretend they're real cities.**

---

**Q127. "Can this work with real Indian cities?"**

**Answer:**

Yes. The network topology is data-driven.

We can replace the demonstration checkpoints and corridors with real geocoded hubs, satellite towns, road networks, rail schedules and operational constraints.

---

**Q128. "Where do your freight costs come from?"**

If they're not documented in README:

The README doesn't specify a real-world cost-data source. In the current prototype, cost values are part of the modeled optimization inputs/data.

For deployment, we'd need carrier tariffs, fuel costs, handling costs, rail freight rates and other operational costs.

**This honesty is much better than inventing a source.**

---

**Q129. "Where does delay probability come from?"**

Same principle:

The current README defines delay probability as the Class B risk input but doesn't describe an ML/data pipeline for estimating it.

In production, it should be derived from historical route and carrier performance, traffic, weather, seasonality and other operational factors.

---

**Q130. "Where is your database?"**

Your README doesn't show a conventional database.

**Answer:**

The current architecture has a repository abstraction and static/demo data. A production deployment can connect that repository layer to a relational or distributed database without changing the core optimization logic.

---

**PART 16 — FAILURE / EDGE CASE QUESTIONS**

**Q131. What if there is no feasible route?**

The optimizer should return no feasible solution rather than silently producing an invalid plan.

The API should communicate that clearly to the user.

---

**Q132. What if two shipments have incompatible temperature requirements?**

They shouldn't be consolidated into the same cold-chain movement unless the shared environment can satisfy both requirements.

---

**Q133. What if one shipment is Class A and another Class B?**

They can potentially share transportation infrastructure if operationally compatible, but their risk calculations remain separate because their loss mechanisms differ.

---

**Q134. What if cargo value is zero?**

The expected loss becomes zero under the given formula, although validation should probably reject nonsensical shipment values depending on business requirements.

---

**Q135. What if delay probability is 1?**

For Class B:

$$ExpectedLoss = CargoValue \times PenaltyRate$$

---

**Q136. What if penalty rate is 0?**

Expected loss is:

0

even if delay probability is high.

---

**Q137. What if penalty rate is greater than 1?**

The README explicitly restricts it to [0,1], so the API validation should reject it.

---

## PART 17 — SOFTWARE ENGINEERING QUESTIONS

### Q138. Why have configuration in config.py?

To centralize configurable settings instead of hardcoding them throughout the application.

---

### Q139. Why structured logging?

README mentions JSON structured logging with UUID tracing.

#### Answer

Structured logs make production debugging and monitoring easier because logs can be searched and correlated programmatically.

UUID tracing allows us to associate logs belonging to the same request.

---

### Q140. Why use UUID tracing?

If a request passes through multiple modules, a unique request ID lets us trace that request across the system.

---

### Q141. Why custom exceptions?

To provide predictable, structured error handling rather than exposing raw internal exceptions to the API client.

---

### Q142. What is CI?

Continuous Integration.

It automatically runs checks such as tests/builds whenever code changes are pushed, helping catch regressions early.

---

## PART 18 — FRONTEND QUESTIONS

### Q143. What does the dashboard show?

From README, the frontend contains:

- tabs
- maps
- badges
- cards

- dashboard state
- shipment information

**Say:**

The frontend acts as the decision-support interface where the user can inspect shipment information, network routes and the generated plans.

---

#### **Q144. Why use maps?**

Logistics is inherently spatial.

A map makes the route, hubs, origin and destination much easier to understand than presenting everything as text.

---

#### **Q145. Why Indian currency formatting?**

The README specifically mentions:

Indian currency ₹L/Cr formatting.

**Answer**

Since the target users are Indian MSME and agricultural logistics operators, displaying financial values using Indian units such as lakh and crore makes the dashboard more natural for them.

---

### **PART 19 — “SHOW ME” QUESTIONS**

Judges may ask you to **open the code**.

Know these files:

**If they ask:**

“Show me risk calculation.”

Go to:

backend/app/engine/risk.py

---

“Show me classification.”

backend/app/engine/classification.py

---

“Show me optimizer.”

backend/app/engine/optimization\_engine.py

---

**“Show me routing.”**

backend/app/engine/road\_route.py

backend/app/engine/rail\_route.py

---

**“Show me API.”**

backend/app/api/routes.py

---

**“Show me schemas.”**

backend/app/api/schemas.py

---

**“Show me frontend API communication.”**

frontend/src/api/client.ts

The README gives this exact directory structure.

---

## **PART 20 — VERY TOUGH QUESTIONS**

Now we're getting into **technical judge territory**.

---

### **Q146. Why isn't your risk model truly Arrhenius?**

Your README calls it:

“Q10/Arrhenius Spoilage”

**If challenged:**

The current implementation uses the Q10 temperature-acceleration formulation.

A strict Arrhenius formulation would explicitly model reaction kinetics using activation energy and absolute temperature:

$$k = Ae^{-E_a/(RT)}$$

So I would distinguish the implemented Q10 approximation from a full Arrhenius kinetic model.

**This is an excellent technical distinction.**

---

### **Q147. Why not use the full Arrhenius equation?**

**Answer**

Q10 provides a simpler and interpretable temperature-acceleration model when product-specific activation-energy parameters aren't available.

A full Arrhenius model would require reliable product-specific kinetic parameters and temperature-dependent degradation data.

---

**Q148. Is shelf life really linear?**

Your current formula effectively assumes baseline shelf-life consumption proportional to time.

**Answer:**

It's a simplified model.

Real degradation can be nonlinear and product-specific.

The current approach provides a computationally simple deterministic approximation, while a production system could replace it with validated degradation curves.

---

**Q149. Is one temperature reading enough?**

**Answer:**

No, not for a high-fidelity cold-chain system.

A time series is much better because thermal exposure is accumulated over time.

---

**Q150. Does lower temperature always mean lower risk?**

**Answer:**

**Don't blindly say yes.**

Not necessarily. Some products have lower-bound storage requirements as well.

Our model specifically defines a safe temperature band, so going outside the permitted lower or upper limit constitutes a breach.

---

**Q151. What is the difference between risk and expected loss?**

**Answer**

**Risk score** describes normalized modeled probability/exposure.

**Expected loss** converts that risk into money.

For example:

$$Risk = 0.2$$

doesn't tell us the financial impact by itself.

But:

$$0.2 \times ₹10,00,000 = ₹2,00,000$$

does.

---

**Q152. Why optimize expected loss instead of risk score?**

**Answer**

Because two shipments with the same risk can have completely different financial consequences.

A 20% risk on ₹10,000 cargo is very different from 20% risk on ₹1 crore cargo.

Expected loss captures this difference.

---

**Q153. Why include cargo value?**

Exactly for the above reason.

---

**Q154. Could the cheapest solution actually be the best?**

**Answer**

Yes, if the additional risk doesn't justify paying more for a faster route.

That's why the system exposes multiple scenarios rather than assuming that lower risk is always worth any additional cost.

---

**Q155. How would you determine  $\alpha$  and  $\beta$  in the real world?**

**Answer**

They shouldn't be arbitrary in production.

They could be determined based on:

- business priorities
- cargo criticality
- historical loss
- SLA requirements
- customer willingness to pay
- risk tolerance

Another approach is to present the Pareto frontier and let the decision-maker select a preferred trade-off.

---

**Q156. What if cargo is extremely valuable?**

**Answer**



Expected loss becomes larger, making risk reduction economically more significant.

Therefore, the optimizer may choose a more expensive but safer route.

---

**Q157. What if cargo is very cheap?**

The optimizer may prefer a cheaper route because the economic consequence of risk is smaller.

---

**Q158. How would you incorporate vehicle capacity?**

**Answer**

Vehicle capacity would become a hard constraint in CP-SAT.

For example:

$$\sum \text{shipment\_weight} \leq \text{vehicle\_capacity}$$

A vehicle cannot be assigned shipments exceeding its capacity.

---

**Q159. How would you incorporate delivery deadlines?**

**Answer**

We can introduce time-window constraints:

$$\text{arrival time} \leq \text{deadline}$$

and CP-SAT can ensure that selected plans satisfy the required time windows.

---

**Q160. How would you handle multiple vehicles?**

Introduce decision variables representing vehicle assignments and enforce:

- capacity
  - route compatibility
  - departure time
  - vehicle availability
  - possibly mode-specific constraints.
- 

**PART 21 — SIH BUSINESS/JUDGE QUESTIONS**

These are less technical.

---

**Q161. What is innovative about this?**

## **Answer**

The innovation isn't simply route finding.

It's the integration of:

**cold-chain risk + economic SLA risk + multimodal consolidation + multi-objective optimization**

into one decision-support system.

---

### **Q162. What's the biggest limitation?**

A strong honest answer:

The current implementation uses a controlled 20-city network and simplified risk inputs.

The next step is integrating real-world dynamic data such as IoT temperature streams, traffic, weather, real rail schedules and historical delay data.

---

### **Q163. What is your future scope?**

Say:

#### **Phase 1**

Real-world network data.

#### **Phase 2**

IoT temperature/GPS integration.

#### **Phase 3**

ML-based delay prediction.

#### **Phase 4**

Dynamic/rolling-horizon optimization.

#### **Phase 5**

Large-scale deployment.

---

### **Q164. How will you monetize it?**

Possible model:

B2B SaaS for logistics operators and MSMEs, potentially with usage-based pricing based on shipment volume or optimization requests.

Could also offer:

- enterprise licensing
- API access

- logistics analytics
  - premium real-time monitoring
- 

#### **Q165. How is this useful to the government?**

Potentially:

Government agricultural supply chains and public-health cold chains could use similar decision-support infrastructure for moving perishable commodities and critical medical supplies efficiently.

Be careful: **don't claim government adoption unless you have it.**

---

#### **Q166. Why MSMEs specifically?**

##### **Answer**

Large logistics companies may already have sophisticated optimization systems, while smaller shippers often lack access to such decision-support capabilities.

A platform that exposes optimized multimodal options through a relatively simple interface could therefore provide value to MSMEs.

---

### **PART 22 — QUESTIONS THEY MAY ASK YOUR TEAM MEMBERS**

This is important.

If your team has different roles, judges might point at one person and ask:

“What did YOU do?”

You need to know the whole project at least conceptually.

They may ask:

#### **Backend person**

What API did you build?

How does /plan work?

How is validation handled?

How does the optimizer receive data?

#### **ML/algorithm person**

Explain Q10.

Explain the objective function.

Why CP-SAT?

What are your constraints?

## Frontend person

How does React communicate with backend?

What happens if API fails?

How is state managed?

## DevOps person

Why Docker?

How does deployment work?

What does CI do?

**Do not let someone know only their tiny section.**

---

## PART 23 — RAPID-FIRE QUESTIONS

You should be able to answer these in **one sentence**.

| Question               | Answer                                                  |
|------------------------|---------------------------------------------------------|
| What is Q10?           | Temperature sensitivity factor.                         |
| What is CP-SAT?        | Constraint Programming + SAT based optimization solver. |
| What is OR-Tools?      | Google's optimization toolkit.                          |
| What is FastAPI?       | Python framework for APIs.                              |
| What is React?         | Frontend component library/framework ecosystem.         |
| What is OSRM?          | Open-source road routing engine.                        |
| What is consolidation? | Combining compatible shipments.                         |
| What is dwell time?    | Time spent waiting/handling rather than moving.         |
| What is p95?           | 95th percentile latency.                                |
| What is Class A?       | Perishable cargo.                                       |
| What is Class B?       | Non-perishable economic-delay cargo.                    |
| Vaccine Q10?           | 2.5.                                                    |
| Grape Q10?             | 2.2.                                                    |
| Vaccine safe range?    | 2–8°C.                                                  |
| Grape safe range?      | 4–12°C.                                                 |

| Question                  | Answer    |
|---------------------------|-----------|
| Vaccine shelf life?       | 48 h.     |
| Grape shelf life?         | 72 h.     |
| Number of cities?         | 20.       |
| Number of hubs?           | 5.        |
| Number of satellites?     | 15.       |
| Number of rail corridors? | 5.        |
| Number of tests?          | 31.       |
| Coverage?                 | 92%.      |
| p50?                      | 46.58 ms. |
| p95?                      | 55.16 ms. |
| Cheapest $\alpha$ ?       | 0.90.     |
| Cheapest $\beta$ ?        | 0.10.     |
| Fastest $\alpha$ ?        | 0.20.     |
| Fastest $\beta$ ?         | 0.80.     |
| Balanced $\alpha$ ?       | 0.55.     |
| Balanced $\beta$ ?        | 0.45.     |

All of these values are directly documented in the README.

---

## PART 24 — THE 10 QUESTIONS I THINK ARE MOST DANGEROUS

If I were sitting on the judging panel, these are the questions I'd use to expose whether the team actually understands the project.

1.

**“Where exactly is the AI?”**

→ Optimization + deterministic decision intelligence; current implementation isn't primarily ML.

2.

**“Why CP-SAT instead of Dijkstra?”**

→ This is a multi-shipment constrained optimization problem, not simply shortest path.

3.

**“Explain Q10 mathematically.”**

→ Know the exponent.

**4.**

**“What happens at 15°C for a vaccine?”**

→ Outside 2–8°C → hard breach → risk 1.0 under current model.

**5.**

**“Is this really Arrhenius?”**

→ Current implementation uses Q10 formulation; full Arrhenius is different.

**6.**

**“How do you get delay probability?”**

→ Current README doesn't specify ML/data source; production should use historical operational data.

**7.**

**“Where do real-time temperatures come from?”**

→ Current sample uses simulated temperature; production should integrate IoT.

**8.**

**“Can this scale beyond 20 cities?”**

→ Hierarchical/decomposed optimization, candidate pruning, rolling horizon.

**9.**

**“Why should I trust your risk score?”**

→ Deterministic, interpretable mathematical model; production requires validation against empirical/product-specific data.

**10.**

**“What is actually innovative here?”**

→ Joint optimization of freight cost, cargo-loss risk, SLA risk, multimodal consolidation and cold-chain constraints.

---

## **PART 25 — YOUR 60-SECOND PITCH**

If they start with:

**“Explain your project.”**

Don't start explaining formulas.

Say this:

Our project is an AI-assisted multimodal logistics optimization platform designed for MSME and agricultural logistics.

The core problem we're solving is that logistics decisions cannot be optimized purely on transportation cost. For perishable goods, factors like temperature, transit time and dwell time can cause spoilage, while for non-perishable goods, delays can cause SLA penalties and financial losses.

So we divide shipments into two categories. For perishable cargo, we use a Q10-based temperature-time degradation model to calculate a normalized risk score and expected cargo loss. For non-perishable cargo, we model expected economic loss from the probability of delay, cargo value and penalty rate.

We then combine freight cost and expected loss into a weighted objective and use Google OR-Tools CP-SAT to find optimized multimodal consolidation plans.

The system models a 20-city network with 5 multimodal hubs and 15 satellite towns, and currently produces three decision options: cheapest, fastest/lowest-risk and balanced.

Our key idea is that instead of asking only "What is the cheapest route?", we're asking "What is the best logistics decision when transportation cost and potential loss are considered together?"

---

## **PART 26 — IF THEY ASK “WHY SHOULD WE SELECT YOUR PROJECT?”**

Use this:

The strength of our project is that we are not treating logistics as a simple shortest-path problem.

We combine multimodal transportation, shipment consolidation, cold-chain risk, economic delay risk and constrained optimization into one decision-support system.

The system is also explainable: instead of producing a black-box recommendation, we can explain why a route was selected in terms of freight cost, transit and dwell time, temperature exposure and expected economic loss.

Our current prototype establishes the optimization framework, and it can be extended with real-time IoT data, historical logistics data and machine-learning-based delay prediction for a production-scale system.

---

## **PART 27 — IF THEY TRY TO ATTACK YOUR PROJECT**

This is perhaps the most important section.

A judge may say:

**“This isn't AI.”**

Don't argue.

Say:

“You're right that the current core is not a conventional ML model. We use deterministic risk modeling and constraint optimization because those are more interpretable for this problem. Our ML

extension would be to learn uncertain variables such as delay probability, ETA and temperature excursion risk from historical data.”

---

They may say:

**“Your temperature is simulated.”**

Say:

“Yes, in the current prototype. The API supports temperature input to demonstrate the risk engine. The production extension is to replace that input with timestamped IoT telemetry.”

---

They may say:

**“20 cities isn't realistic for India.”**

Say:

“The 20-city network is a controlled benchmark topology. The architecture separates network data from the optimization logic, so the same framework can be populated with real Indian network data.”

---

They may say:

**“Your Q10 model is oversimplified.”**

Say:

“Yes, it is a simplified deterministic model. Its advantage is interpretability and low computational overhead. For deployment, we'd calibrate it using product-specific validated degradation data.”

---

They may say:

**“Your delay probability isn't predicted.”**

Say:

“Correct. The current implementation treats delay probability as an input to the economic-risk model. Our next layer would learn that probability from historical shipment, traffic, weather and carrier-performance data.”

---

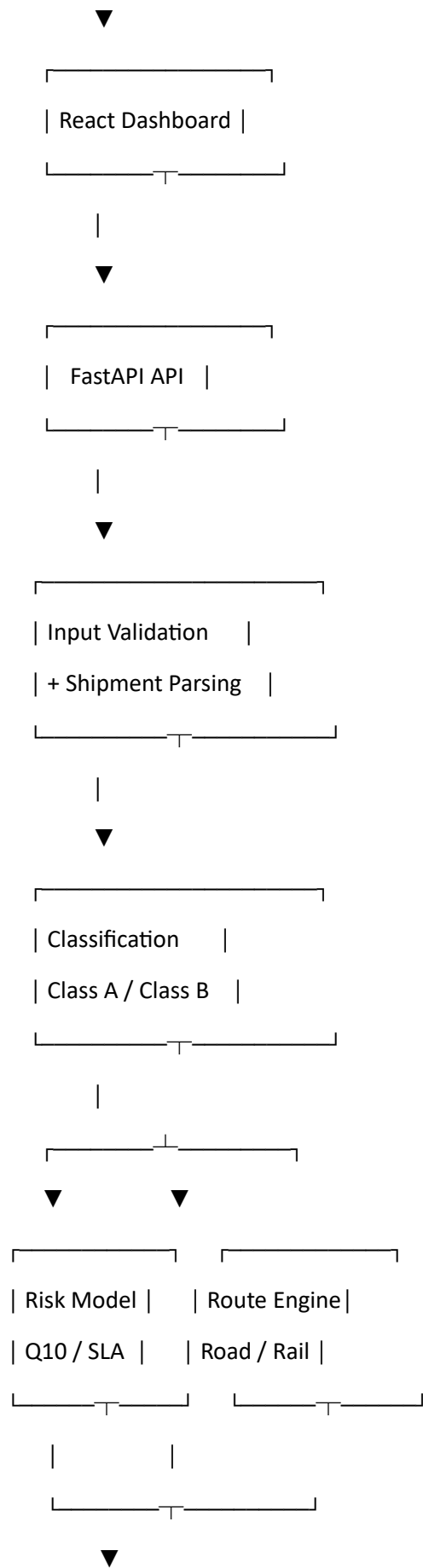
## **PART 28 — THE ARCHITECTURE YOU SHOULD HAVE IN YOUR HEAD**

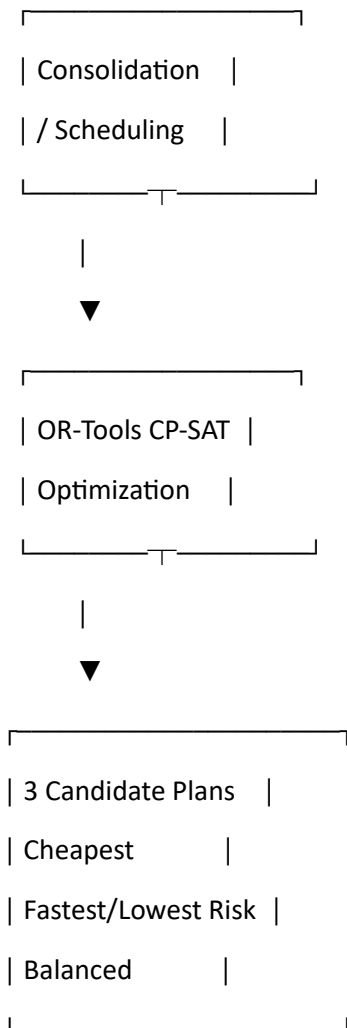
Memorize this:

USER

|







Your README's backend structure supports essentially this separation of responsibilities.

---

## PART 29 — ONE THING I REALLY WANT YOU TO FIX BEFORE THE ROUND

Don't memorize the README like a parrot.

Understand these **five relationships**:

1.

Temperature ↑ → degradation factor ↑ → risk ↑ → expected loss ↑

2.

Transit/dwell ↑ → shelf-life consumption ↑ → risk ↑

3.

Cargo value ↑ → expected loss ↑

4.

Delay probability ↑ → Class B expected loss ↑

5.

**Risk weight  $\beta \uparrow \rightarrow$  optimizer becomes more willing to spend money to reduce risk**

If you understand those five chains, you can answer a huge number of questions even if the judge phrases them differently.

---

## **FINAL 15-MINUTE REVISION SHEET**

Before walking into the room, make sure you can answer these **without looking anywhere**:

### **Project**

- What problem?
- Who uses it?
- Why is it important?
- What's innovative?

### **Math**

- Q10
- Hard breach
- Fraction consumed
- Risk score
- Expected loss
- Class B formula
- $\alpha/\beta$  objective

### **Optimization**

- What is CP-SAT?
- Why CP-SAT?
- What are constraints?
- Why not Dijkstra?
- What is Pareto optimality?

### **Architecture**

- React
- FastAPI
- OR-Tools
- OSRM

- Risk engine
- Orchestrator
- Repository

## Numbers

- 20 cities
- 5 hubs
- 15 satellites
- 5 rail corridors
- 31 tests
- 92% coverage
- p50 46.58 ms
- p95 55.16 ms
- vaccine Q10 = 2.5
- grapes Q10 = 2.2
- vaccine = 2–8°C
- grapes = 4–12°C

## Most important conceptual answer:

**“Our current system uses deterministic physics-based risk modeling and constraint optimization rather than conventional ML. The next ML layer would predict uncertain quantities such as delay probability and ETA from historical and real-time data.”**

If you can explain **that sentence confidently**, you will be much safer when the judges start probing the “AI” part.

And one final thing: **don't claim features that aren't actually implemented**. SIH judges often test this by asking you to open the relevant file or modify an input live. Your README explicitly documents what is implemented, so stay within that boundary and frame the rest as future scope.